Few-Shot Pulmonary Vessel Segmentation Based on Tubular-Aware Prompt-Tuning

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***Abstract*— Segmentation of the pulmonary vessel from computed tomography (CT) images plays a crucial role in the diagnosis and treatment of various lung diseases. Although deep learning-based approaches have shown remarkable progress in recent years, their performance is often hindered by the lack of high-quality annotated datasets, in which the complex anatomy and morphology of pulmonary vessels make manual annotation challenging, time-consuming, and prone to errors. To address this, we propose PV25, the first dataset that features finely paired annotations of both pulmonary vessels and airways. Moreover, we propose TPNet, a novel tubular-aware prompt-tuning framework for pulmonary vessel segmentation under few-shot training with limited annotations. Specifically, based on an advanced and frozen segmentation backbone, TPNet proposes tunable encoding and decoding networks that learn tubular structures as transfer learning priors, bridging the gap between the source and target pulmonary vessel domains. Specifically, TPNet is built in an encoder-decoder manner, including the fixed segmentation backbone, tunable encoding and decoding networks. In encoding stage, the Morphology-Driven Region Growing (MDRG) module is developed to leverage the tubular connectivity of vessels to guide the network in capturing fine-grained features of pulmonary vessels. In decoding stage, the Cross-Correlation Guidance (CCG) module is introduced to integrate multi-scale correlations between airway and vessel structures in a coarse-to-fine manner. Extensive experiments conducted on multiple datasets demonstrate that TPNet achieves state-of-the-art performance in pulmonary vessel segmentation under limited training data. Besides, TPNet shows strong performance in related tasks such as airway segmentation and artery-vein classification, highlighting its robustness and versatility.**

***Index Terms*— Pulmonary vessel segmentation, prompt tuning, transfer learning, tubular-aware.**

I. Introduction

COMPUTED tomography (CT) based pulmonary vessel segmentation is crucial for the diagnosis and treatment of diseases such as pulmonary embolism and chronic obstructive pulmonary disease [1], which is becoming the third leading cause of death worldwide, with an estimated annual mortality of approximately 3.2 million deaths [2]. The past decade has witnessed the increasing research interests of applying deep neural network (DNN) [3] for pulmonary vessel segmentation [4], [5]. However, DNNs usually require large-scale training data with high-quality annotations to guarantee the performance and generalization capability for medical imaging processing [6]. For pulmonary vessel segmentation, the complex anatomical structure of pulmonary vessels makes manual annotation time-consuming, labor-intensive, and reliant on extensive clinical expertise. For instance, even for an experienced radiologist, the high-quality vessel annotation process of a single CT scan requires approximately 6 to 15 hours [7], which is about 60 to 150 times longer than the time required for annotating the lung parenchyma [8]. As far as we know, there only exists two public pulmonary vessel datasets [9], [10], with less than 60 annotated cases. The shortage of both quantity and quality of available annotations limits the effectiveness and practical applicability of existing vessel segmentation methods in clinical scenarios. Even worse, DNN based methods typically require retraining when applied to CT scans from different centers and devices [11]. Therefore, it is of crucial importance to develop a few-shot learning framework for pulmonary vessel segmentation, enabling efficiently transfer the prior knowledge across domains.

However, as shown in Fig. 1-(a), even applying with the transfer learning technique, existing vessel segmentation methods, such as UNETR [12] and nnFormer [13], still fail to accurately segment the vessels when given the limited available scans. That is mainly because the segmentation methods struggle to learn the complete vessel structures when training

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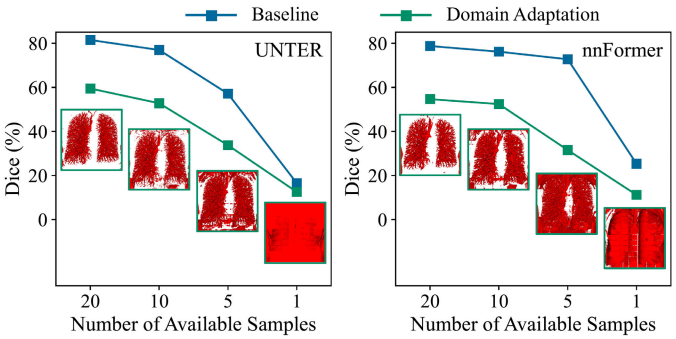
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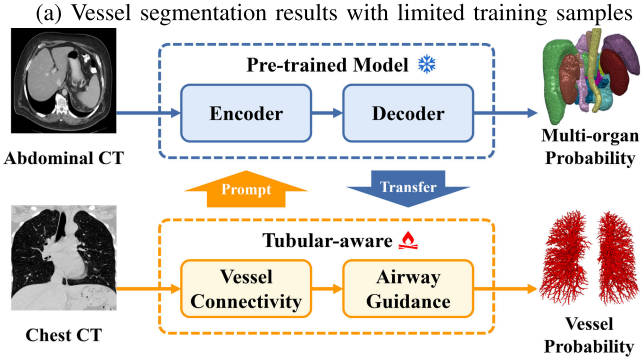
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(a) Vessel segmentation results with limited training samples



(b) Framework of our tubular-aware prompt-tuning network

Fig. 1. (a) Existing pulmonary vessel segmentation methods with transfer learning technique (domain adaptation), exhibit significant performance degradation in few-shot settings due to their inability to model complete vessel structures from limited training data. (b) This paper proposes a novel tubular-aware prompt-tuning network for pulmonary vessel segmentation under few-shot settings, by leveraging the tubular/segmentation priors from airway/multi-organ annotations.

data is insufficient. Although transfer learning technique can somehow mitigate the data scarcity issue, the complicated characteristics of pulmonary vessel segmentation make it challenging to effectively adapt knowledge from other medical imaging tasks. As shown in the figure, the anatomical structure of pulmonary vessels is highly intricate, forming a hierarchically organized tubular network with extensive branching. The smaller peripheral vessels are fine and numerous, making them susceptible to noise interference from surrounding tissues, while the larger central vessels are closely adjacent to structures such as airways and lymphatic vessels, often exhibiting blurred boundaries. This complex spatial distribution pattern exacerbates the difficulty of learning complete vessel structures with limited scans in few-shot learning settings.

To address this challenge, we set up the first attempt to leverage the inherent structural information of pulmonary vessels, in order to construct the homologous vessel space and learn the meta information for the few-shot scenarios. Different from other medical image segmentation tasks, pulmonary vessels exhibit two prominent characteristics: 1) strong topological connectivity and 2) anatomical correlation with airways, both of which provide potential supervisory priors. On the one hand, pulmonary vessels inherently possess topological connectivity, indicating that local vessel features are highly constrained by the global structure [14]. Therefore, it has the potential to explicitly incorporate vessel topology priors into the segmentation method, as the way to segment the vessel details in few-shot learning. On the other hand, the morphology of pulmonary vessels is highly correlated with airway structures in an anatomical context. These two structures not only exhibit spatial co-localization but also share similar branching patterns [15]. For instance, pulmonary arteries typically extend along the bronchial tree and follow its hierarchical structure [16]. Moreover, airways can be regarded as a coarse representation of pulmonary vessels, being able to provide morphological prior for vessel segmentation [17].

In this paper, we propose an innovative framework to tackle the challenge of pulmonary vessel segmentation under few-shot conditions. First, we introduce the PV25 dataset, which comprises 25 CT scans with both annotated pulmonary vessels and airways, providing a new and reliable benchmark for few-shot learning. More importantly, as illustrated in Fig. 1-(b), we propose a tubular-aware prompt-tuning network (TPNet) in an encoder-decoder manner, by learning the anatomical structures as transfer learning priors. In the encoding stage, a morphology-driven region growing (MDRG) module is developed to exploit vessel connectivity to facilitate the extraction of complete vessel structures. Besides, we develop a lightweight adapter to improve the network's adaptability to different data distributions. In the decoding stage, a dual-path structure is developed in TPNet, to separately segment pulmonary vessels and airways. This architecture explicitly enables the network to utilize coarse multi-scale airway information when segmenting vessels. Furthermore, we introduce a cross-correlation guided (CCG) module to establish multi-scale associations between vessels and airways during decoding, progressively refining vessel feature representations and ultimately enhancing segmentation accuracy. It is worth noting that, the proposed tubular-aware MDRG and CCG modules are supposed to improve the vessel segmentation performance, especially for he few-shot scenario, by leveraging the additional anatomical priors from the similar domain. The code and dataset are publicly available at https://github.com/Hoodini-Gzj/PTNet. The main contributions of this study can be summarized as follows.

- We introduce a new dataset, PV25, the first to provide finely annotated paired vessel and airway labels for 25 high-resolution chest CT scans.
- We propose a novel tubular-aware prompt-tuning framework, namely TPNet, to achieve a few-shot segmentation of the pulmonary vessels, via using the prior knowledge of tubular structures.
- We develop new MDRG and CCG modules in the TPNet, to implicitly extract tubular structures and explicitly learn the multi-scale vessel representation from the pulmonary airway, respectively.

II. Related Work

A. Pulmonary Vessel Segmentation

Pulmonary vessel segmentation is crucial for diagnosing and managing pulmonary diseases such as embolism, hypertension, and COPD. Early methods, relying on handcrafted features or model-based techniques, struggled with intensity variations and anatomical complexity, limiting robustness

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and generalization. Deep learning, however, has significantly improved vessel morphology capture. For instance, Gu et al. [18] proposed a cascaded network to segment high-intensity structures and differentiate vessels from non-vessels. Nam et al. [19] developed a deep learning-based non-contrast CT segmentation algorithm for evaluating small-vessel density loss in COPD patients. Wu et al. [4] introduced a multi-scale interactive UNet, enhancing microvessel segmentation accuracy through multi-scale feature extraction and parameter sharing. Chen et al. [20] employed a learnable Gabor convolutional kernel for efficient brain and pulmonary vessel segmentation. Wu et al. [5] designed a Transformer-based network with a channel-enhanced attention module for segmentation and artery-vein separation in both non-contrast and contrast-enhanced CT images. Despite these advances, most methods still rely on large annotated datasets, which are difficult to obtain due to the intricate topology and subtle contrast of thin vessels. Consequently, the reliance on small or partially labeled datasets restricts model generalization, highlighting the need for data-efficient approaches for effective vessel segmentation with limited annotations.

B. Few-Shot Segmentation

Few-shot segmentation aims to learn effective representations from limited annotated samples, which is particularly useful in medical imaging, where expert annotation is costly and time-consuming. Recently, parameter-efficient fine-tuning techniques have emerged as promising solutions, allowing large pre-trained networks to adapt to new domains with minimal parameter adjustments. These techniques have been extended to various medical image analysis tasks. For instance, Du et al. [21] integrated adapters into transformer layers to adjust image representations in the backbone and employed a prompt generator to further guide skin lesion segmentation under limited data conditions. Huang et al. [22] introduced grouped prompts to share embeddings and prompt tokens at low adjustment cost, achieving strong performance across three datasets. Ye et al. [23] proposed a universal segmentation model named UniSeg, which uses learnable universal prompts to capture correlations across multiple tasks and convert them into task-specific prompts. Additionally, domain adaptation has been explored to address domain shifts and mitigate data scarcity. Li et al. [24] utilized prototype embeddings mapped to the target domain, fine-tuning only projection layers to reduce overfitting. Keaton et al. [25] used a contrastive loss to adapt pre-trained models to target domains. Despite these advances, challenges persist in vessel segmentation tasks, where prompt-based methods may not capture fine tubular connectivity and branching hierarchies effectively, and domain adaptation methods face limitations when source domains are structurally dissimilar or unavailable [26].

C. Prior-Guided Segmentation

In recent years, prior-guided approaches have emerged as a powerful strategy for medical image segmentation, effectively leveraging domain-specific structural information to enhance model performance. For example, Wu et al. [27] utilized prototype and contextual contrasts, extracting anatomical background from non-target images as negative samples and obtaining positive samples from support images, thereby improving feature discriminability and data efficiency for abdominal organ segmentation. Lin et al. [28] employed foreground masks from support images as explicit priors, guiding attention mechanisms toward relevant anatomical regions, and iteratively refining segmentation accuracy for abdominal and cardiac images. Xiao et al. [29] proposed a framework for skin lesion segmentation, using co-occurrence regions as structural prior masks to reduce segmentation bias. Shao et al. [30] introduced a gradient-based loss to encode edge and connectivity priors for retinal vessels, leveraging a two-stage network to preserve tubular topology with limited training samples. Wang et al. [31] integrated vascular topological priors into a retinal vessel segmentation network to improve continuity modeling. Collectively, these studies illustrate the critical role of anatomical and morphological priors in enhancing the accuracy and robustness of few-shot segmentation, motivating our tubular-aware approach for pulmonary vessel segmentation under data-scarce scenarios.

III. Database and Analysis

A. Database Establishment

The PV25 database in this study is built upon the public CT database, ATM22 [32], for airway segmentation. From ATM22, we selected 25 high-resolution chest CT scans from healthy individuals. Each CT scan comprises a varying number of slices, ranging from 500 to 741, with slice thicknesses between 0.450mm and 1.000mm. All slices have an axial dimension of 512×512 pixels and a spatial resolution between 0.500mm and 0.820mm. We initially truncated the voxel intensities to the range of [−950, +250] Hounsfield units. Based on this, thresholding method and Frangi filter were respectively applied to segment the lung parenchyma region and enhance vessel structures. Two experienced radiologists refined the segmentation masks based on anatomical priors and existing airway annotations. Vessel annotations with less than 50% overlap consistency were sent for further refinement by another senior radiologist. This process was repeated until no discrepancies remained, and the final annotations were double-checked by the senior radiologist. As shown in Fig. 2 (a), the airway and vessel are annotated in blue and red, respectively. Note that this retrospective study adhered to the principles outlined in the World Medical Association Declaration of Helsinki, with all data anonymized to ensure privacy.

B. Database Analysis

Based on the established PV25 database, the data analysis is conducted with the following findings.

Finding 1: *There is a great similarity between pulmonary airways and vessels.*

Analysis: As claimed in the existing study [7], [16], the pulmonary airways and vessels share great similarities in CT images. Specifically, both the airway and vessel structures exhibit a branching tree-shaped morphology that extends from the hilum to the alveoli, forming complex networks.

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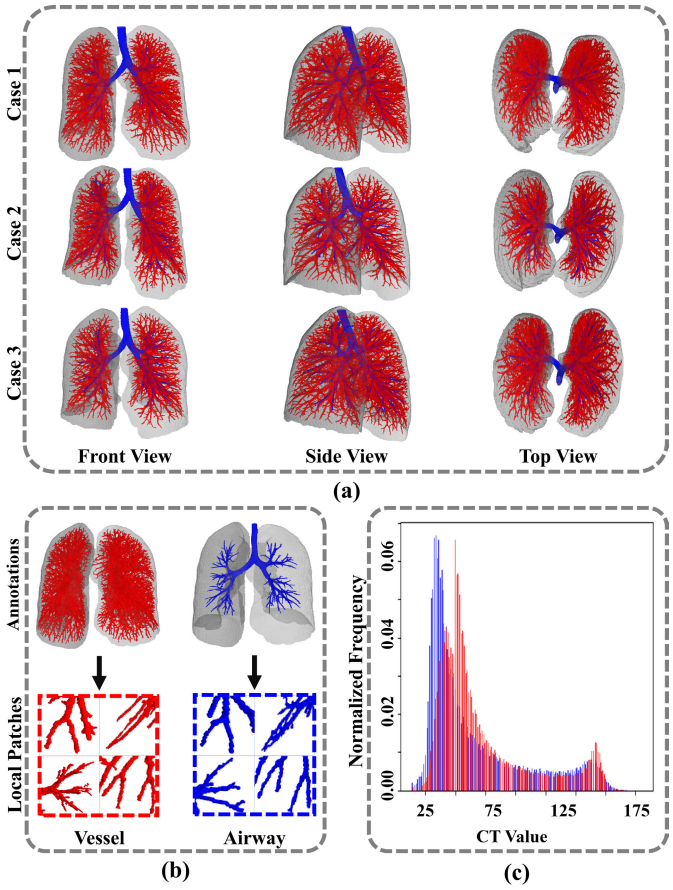


Fig. 2. Illustration and statistical analysis of PV25. In figures, the pulmonary vessels and airways are annotated in red and blue, respectively. (a) Three 3D chest CT scan examples with vessels and airways annotations, in the order of front, side, and top views. (b) A patch view highlighting the local morphological similarities between vessels and airways. (c) CT value distributions for the vessels and walls of airways.

As illustrated in Fig. 2 (b), this similarity becomes obvious when zooming in the terminal points and bifurcations. Based on this, we further calculate the CT intensity values of airways and vessels over our PV25 database, comparing their distributions in Fig. 2 (c). The correlation coefficient (CC) and Kullback-Leibler divergence (KL) between these two distributions are 0.72 and 0.18, indicating a high similarity of CT intensity. The above results conclude the analysis of *Finding 1*.

Finding 2: *The data scale significantly affects the performance of pulmonary vessel segmentation, even with transfer learning strategies.*

Analysis: Here, to evaluate the pulmonary vessel segmentation over the practical few-shot learning scenario, we compare the performance of two widely-used segmentation models, UNETR [12] and nnFormer [13], given the limited training scans. Note that the training samples are randomly selected. As shown in Table I, the segmentation accuracy obviously degrades when the number of training scans is reduced from 20 to 1. For instance, given 20 training scans, UNETR achieves Dice Similarity Coefficient (DSC) of 81.47, but fails to segment the vessel as the number of scans decreases to 5 and 1, with Dice significantly dropping to 57.10, and 16.55. Furthermore, we compared several existing transfer learning strategies, such as fine-tuning [33], domain adaptation [34], linear probing [35] and prompt-tuning [21]. However, as tabulated in Table I, the segmentation accuracy is still far from satisfactory when given the limited training data. Similar results can be found in nnFormer. The above analysis indicates the challenge of achieving reliable pulmonary vessel segmentation in few-shot scenario, concluding the analysis of *Finding 2*.

TABLE I
DSC PERFORMANCE OF DIFFERENT PULMONARY VESSEL SEGMENTATION METHODS UNDER VARIOUS TRAINING SCAN QUANTITIES

Method	20 Scans	10 Scans	5 Scans	1 Scan
Fully-trained [12]	81.47/78.73	76.94/76.17	57.10/62.74	16.55/25.36
Fine-tuning [33]	83.47/81.64	82.61/80.11	76.24/74.57	37.50/35.91
Domain Adaptation [34]	59.45/54.63	52.78/52.42	33.75/31.56	12.56/11.24
Linear Probing [35]	75.78/74.11	66.61/64.78	54.21/51.94	24.69/20.18
Prompt-tuning [21]	66.34/62.47	65.64/62.41	64.33/61.79	52.55/49.88

IV. METHODOLOGY

A. Method Overview

As shown in Fig. 3, the proposed TPNet features an encoder-decoder architecture with a morphology encoder, vessel-airway dual-branch decoder, and a pre-trained segmentation backbone. In the encoding stage, a frozen state-of-the-art UNETR model [12] with lightweight adapters is used for feature representation, while the morphology encoder, enhanced with MDRG modules, extracts and integrates tubular features from input scans. In the decoding stage, a dual-branch decoder simultaneously segments the vessel and airway structures, with the CCG module enabling multi-scale information exchange between them. This coarse-to-fine process leverages airway tubular information to refine pulmonary vessel segmentation. During training, we freeze the segmentation backbone and fine-tune the proposed lightweight components, to achieve the quick domain adaptation when given the limited training data. It is important to note that our approach does not fall under the category of source-free domain adaptation. The source domain is abdominal multi-organ segmentation, while the target domain is pulmonary vessel segmentation.

B. Encoding Stage

As shown in Fig. 3, the encoding stage consists of two distinct paths. Given the input CT images, the MDRG encoder is proposed to produce the morphology features by learning the continuity of the tubular structures of pulmonary vessels. Besides, the CT images are also input to a pre-trained UNETR backbone with a developed lightweight adapter, for generating the task-specific features. Then, the encoded features are passed to the following decoding stage.

1) Morphology Encoder: Inspired by the dynamic convolution layers in [36], a set of MDRG modules is developed to preserve the topology of pulmonary vessels. The structure of MDRG is illustrated in Fig. 4. As shown, with the input of CT scans $\mathbf{I}$, the topology features are extracted by three parallel convolutional layers, denoted as Conv_x, Conv_y, and

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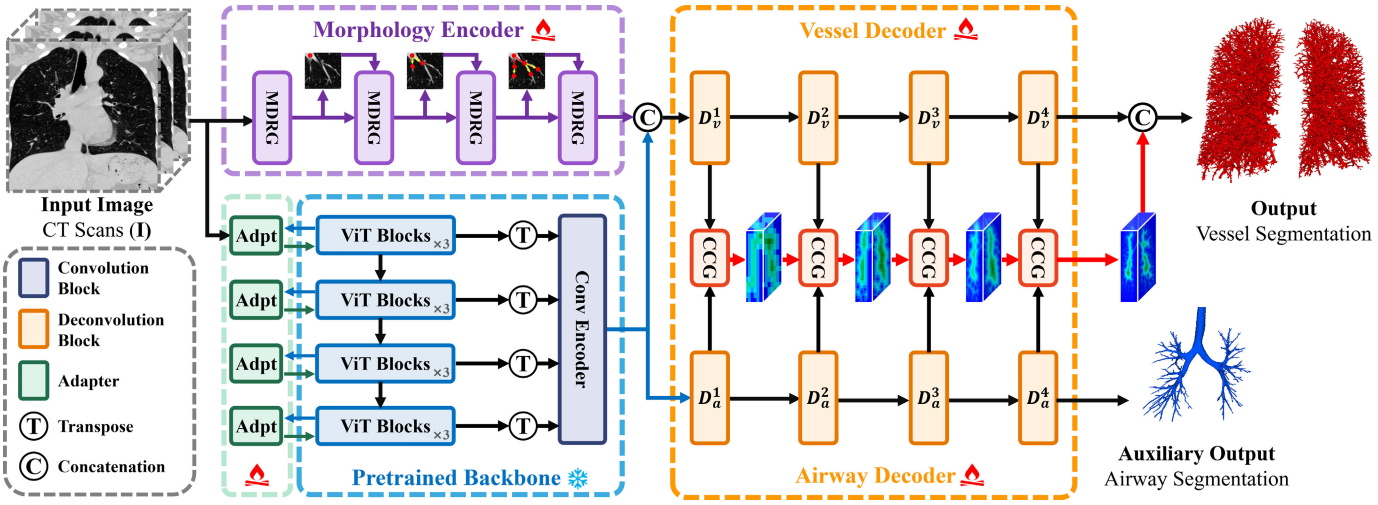


Fig. 3. The proposed TPNet framework for segmenting pulmonary vessels and airways in few-shot learning scenarios, consisting of the pretrained backbone network, morphology encoder, vessel decoder and airway decoder. Note that MDRG and CCG are the developed morphology-driven region growing module and cross-correlation guidance module. The flame and snowflake symbols refer to the tunable and frozen components, respectively.

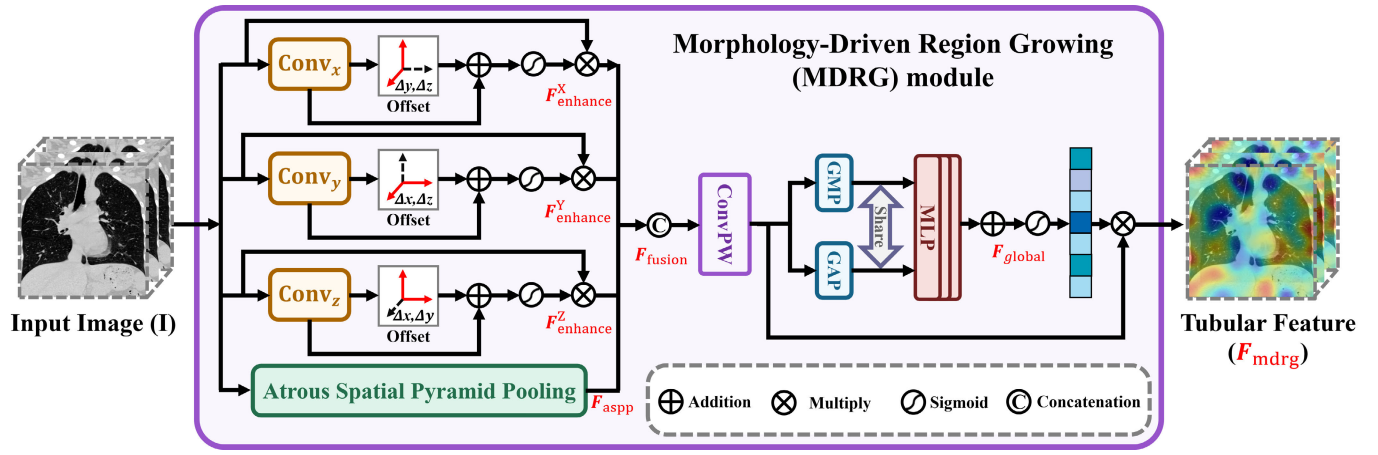


Fig. 4. The detailed illustration of our MDRG module. Note that the intermediate features are noted in red.

Conv_z. Each convolutional layer is enhanced by introducing sequential, learnable offsets to capture the continuous trajectory of the vessel tree in three-dimensional space. Consider a convolutional layer with a kernel size of $7 \times 7 \times 7$ as an example. By applying small displacements Δ_x, Δ_y and Δ_z along the X, Y and Z axes, the receptive field can be adaptively adjusted along all three axes. Each grid position within the kernel is determined incrementally in a cumulative process, where the displacement for each subsequent grid position is offset by a small value $\Delta = \{\lambda | \lambda \in [-1, 1]\}$, relative to the previous grid position, beginning from the central position p. The range of λ is constrained to ensure smooth transitions and prevent large deviations. This iterative adjustment ensures that the receptive field aligns with vessel connectivity, while the offset values are optimized within the defined range to maximize alignment accuracy. For non-integer offsets, bilinear interpolation is applied. Consequently, the adjusted receptive fields $\boldsymbol{R}_x$, $\boldsymbol{R}_y$ and $\boldsymbol{R}_z$ for Conv_x, Conv_y and Conv_z are expressed as:

$$\boldsymbol{R}_x = x_{p\pm c}, y_p + \sum_{p-c}^{p+c} \Delta_y, z_p + \sum_{p-c}^{p+c} \Delta_z$$

$$\boldsymbol{R}_y = x_p + \sum_{p-c}^{p+c} \Delta_x, y_{p\pm c}, z_p + \sum_{p-c}^{p+c} \Delta_z$$

$$\boldsymbol{R}_z = x_p + \sum_{p-c}^{p+c} \Delta_x, y_p + \sum_{p-c}^{p+c} \Delta_y, z_{p\pm c} \quad (1)$$

where p represents the center of the kernel, and $c \in \{0, 1, 2, 3\}$ denotes the distance from the center. Δ_x, Δ_y and Δ_z are the offsets along the X, Y and Z axes, respectively. Through this adjustment, Conv_x captures the tubular response along the X axis, and a soft attention map is generated using a Sigmoid function. The attention map is subsequently applied to the input $\mathbf{I}$, resulting in an enhanced tubular voxel representation along the X axis, denoted as $\boldsymbol{F}^{\text{X}}_{\text{enhance}}$.The representation of $\boldsymbol{F}^{\text{X}}_{\text{enhance}}$ is given as follows:

$$\boldsymbol{F}^{\text{X}}_{\text{enhance}} = \text{Sigmoid}(\text{Conv}_x(\mathbf{I})) * \mathbf{I} \quad (2)$$

Similarly, Conv_y and Conv_z are employed to enhance tubular features along the Y and Z axes, respectively, yielding $\boldsymbol{F}^{\text{Y}}_{\text{enhance}}$ and $\boldsymbol{F}^{\text{Z}}_{\text{enhance}}$. These operations effectively improve the network's capability to capture and refine tubular structures.

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Furthermore, in order to capture the local features for vessel segmentation, an Atrous Spatial Pyramid Pooling (ASPP) module is employed in our MDRG module for generating multi-scale features $\boldsymbol{F}_{\text{aspp}}$. Given the above features ($\boldsymbol{F}^{\text{X}}_{\text{enhance}}$, $\boldsymbol{F}^{\text{Y}}_{\text{enhance}}$, $\boldsymbol{F}^{\text{Z}}_{\text{enhance}}$, and $\boldsymbol{F}_{\text{aspp}}$), a channel attention mechnism is developed to fuse and refine the features:

$$\begin{aligned}
&\boldsymbol{F}_{\text{mdrg}} = \text{Sigmoid}(\boldsymbol{F}_{\text{global}}) \odot \boldsymbol{F}_{\text{fusion}}, \quad \text{where} \\
&\boldsymbol{F}_{\text{fusion}} = \text{ConvPW}(\text{Concat}(\boldsymbol{F}^{\text{X}}_{\text{enhance}}, \boldsymbol{F}^{\text{Y}}_{\text{enhance}}, \boldsymbol{F}^{\text{Z}}_{\text{enhance}}, \boldsymbol{F}_{\text{aspp}})) \\
&\boldsymbol{F}_{\text{global}} = \text{MLP}(\text{GMP}(\boldsymbol{F}_{\text{fusion}})) + \text{MLP}(\text{GAP}(\boldsymbol{F}_{\text{fusion}}))
\end{aligned} \tag{3}$$

In above equation, ConvPW and Concat denote the point-wise convolution and concatenation operation along the channel dimension. Besides, GMP and GAP are global max pooling and global average pooling for capturing global information, while MLP indicates a shared multi-layer perceptron. Finally, the tubular feature $\boldsymbol{F}_{\text{mdrg}}$ can be obtained.

2) Pretrained Backbone With Lightweight Adapters: Our method leverages a pre-trained UNETR backbone, a medical image segmentation architecture combining ViTs and CNNs. The pre-trained weights were obtained from training on the BTCV abdominal multi-organ segmentation dataset and were directly adopted for transfer learning in our study without additional self-training.[1] The use of UNETR and its pre-trained weights is driven by the fact that abdominal CT contains rich vessel and boundary information, which aids in recognizing pulmonary vessel structures [37]. UNETR splits input scan into non-overlapping patches, projects them into embedding vectors, and adds positional encodings to form a sequential representation for the ViT encoder. During encoding, the ViT's multi-head self-attention captures long-range dependencies and global context, overcoming the limited receptive field of CNNs. In decoding, a symmetric U-shaped architecture combines transformer representations with convolutional upsampling, integrating global semantic features and local spatial details for improved segmentation accuracy. However, the pre-trained weights are not directly transferable due to the disparity in feature representations between abdominal organs and pulmonary vessels. To address this, we introduce a lightweight adapter that adjusts the pre-trained ViT backbone for pulmonary vessel segmentation. The adapter, comprising two MLP layers and a cross-attention mechanism, is integrated at the beginning of the ViT encoder. The input image is first processed into an intermediate representation via the MLP, which is then used as the query for cross-attention with the pre-trained ViT feature vectors (keys and values). During training, the parameters of the lightweight adapter are optimized using the Adam optimizer, with an initial learning rate set to 1×10^{-4}. The optimization process focuses solely on the two MLP layers and the cross-attention mechanism within the adapter, while the weights of the ViT backbone are kept frozen to ensure efficient parameter utilization.

C. Decoding Stage

In the decoding phase, as shown in Fig. 3, two structurally identical decoding paths are developed to predict the segmentation of pulmonary vessels and airways, respectively. Each decoding path consists of four blocks, which are composed of convolutional layers, batch normalization layers, and ReLU activation functions. Furthermore, we introduced the CCG module between each pair of decoders to capture multi-scale correlated features between the pulmonary vessels and airways, as illustrated in Fig. 5. These multi-level features are used for progressively refining the pulmonary vessel segmentation results.

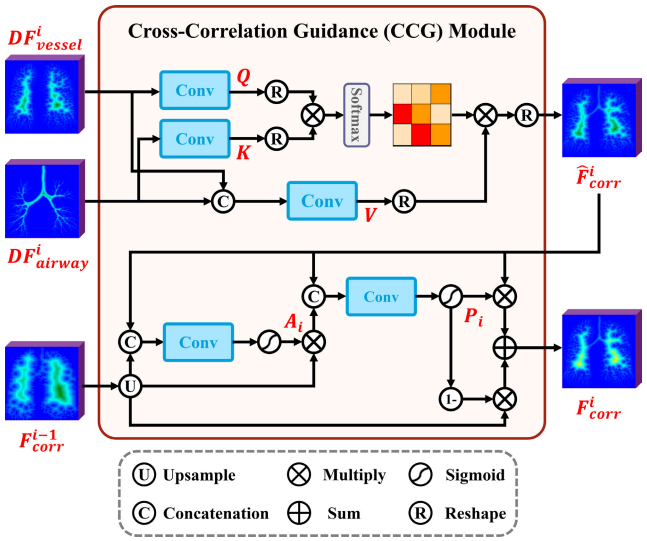


Fig. 5. The detailed structure of our CCG module. As shown, each CCG module leverages the decoding features of vessel and airway paths, to progressively refine the correlated features between two paths.

1) Cross-Correlation Guided Module: For the i-th decoding layer, The CCG module first computes the shallow correlation features $\widehat{\boldsymbol{F}}^i_{\text{corr}}$ between the vessel decoding features $\boldsymbol{DF}^i_{\text{vessel}}$ and the airway decoding features $\boldsymbol{DF}^i_{\text{airway}}$ upon a cross-attention mechanism, as illustrated in the upper part of Fig. 5. This process can be formulated as:

$$\widehat{\boldsymbol{F}}^i_{\text{corr}} = \text{Reshape}(\text{CrossAttention}(\mathbf{Q}, \mathbf{K}, \mathbf{V})) \tag{4}$$

In the above equation, $\mathbf{Q}$ and $\mathbf{K}$ denote the vessel features $\boldsymbol{DF}^i_{\text{vessel}}$ and the airway features $\boldsymbol{DF}^i_{\text{airway}}$, respectively, while $\mathbf{V}$ represents the fused features of $\boldsymbol{DF}^i_{\text{vessel}}$ and $\boldsymbol{DF}^i_{\text{airway}}$. Reshape$(\cdot)$ denotes a reshaping operation used to align the dimensions of the feature maps, while CrossAttention$(\cdot)$ indicates the cross attention block. Through this process, the correlation between pulmonary vessels and airways is captured at each decoding layer and progressively refined in subsequent steps. To learn the comprehensive representation across different levels of correlation between the airway and vessel, adjacent CCG modules are designed to interact with each other. Specifically, the prior correlation features $\boldsymbol{F}^{i-1}_{\text{corr}}$ from the $(i-1)$-th layer and the shallow correlation features $\widehat{\boldsymbol{F}}^i_{\text{corr}}$ from the current layer are jointly utilized to generate the refined correlation features $\boldsymbol{F}^i_{\text{corr}}$, which serve as prior guidance for the next layer. The above process can be formulated as below:

$$\boldsymbol{A}_i = \text{Sigmoid}(\text{Conv}(\text{Concat}(\boldsymbol{F}^{i-1}_{\text{corr}}, \text{Upsample}(\widehat{\boldsymbol{F}}^i_{\text{corr}})))), \tag{5}$$

[1] https://github.com/Project-MONAI/research-contributions/tree/main/UNETR

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where $\boldsymbol{A}_i$ quantifies which information in $\boldsymbol{F}_{\text{corr}}^{i-1}$ can be used to optimize $\widehat{\boldsymbol{F}}_{\text{corr}}^{i}$, establishing an initial connection between adjacent CCG layers. The Sigmoid activation constrains $\boldsymbol{A}_i$ to the range $[0, 1]$, functioning as a gating mechanism. Meanwhile, Upsample$(\cdot)$ aligns the spatial resolution of $\boldsymbol{F}_{\text{corr}}^{i-1}$ with that of $\widehat{\boldsymbol{F}}_{\text{corr}}^{i}$, as the resolution of correlation features progressively increases from lower to higher decoding layers. The dependency between $\boldsymbol{F}_{\text{corr}}^{i-1}$ and $\widehat{\boldsymbol{F}}_{\text{corr}}^{i}$ is then computed as

$$\boldsymbol{P}_i = \text{Sigmoid}(\text{Conv}(\text{Concat}(\widehat{\boldsymbol{F}}_{\text{corr}}^{i}, \boldsymbol{A}_i \cdot \boldsymbol{F}_{\text{corr}}^{i-1}))), \tag{6}$$

where $\boldsymbol{P}_i$ evaluates the contribution of $\boldsymbol{F}_{\text{corr}}^{i-1}$ to $\widehat{\boldsymbol{F}}_{\text{corr}}^{i}$, balancing the relative importance between prior and shallow correlation features at the current level. Subsequently, the correlation features from adjacent layers are dynamically filtered and fused to generate the final output of the CCG module, which serves as the refined correlation features $\boldsymbol{F}_{\text{corr}}^{i}$ for the subsequent layer, as formulated below:

$$\boldsymbol{F}_{\text{corr}}^{i} = (1 - \boldsymbol{P}_i) \cdot \boldsymbol{F}_{\text{corr}}^{i-1} + \boldsymbol{P}_i \cdot \widehat{\boldsymbol{F}}_{\text{corr}}^{i} \tag{7}$$

Through the proposed CCG module, the correlation between the vessel and airway at different semantic levels is dynamically integrated, thereby enhancing the accuracy of pulmonary vessel segmentation. Notably, when $i = 1$, there is no available prior correlation information, and thus $\boldsymbol{F}_{\text{corr}}^{0}$ is undefined. In this case, the shallow correlation features $\widehat{\boldsymbol{F}}_{\text{corr}}^{1}$ are directly used for optimization in the subsequent layer.

2) Loss Function: In this study, we incorporate the DSC loss, $\mathcal{L}_{\text{DSC}}$, to promote accurate segmentation of vessels and airways. The DSC loss is formulated as:

$$\mathcal{L}_{\text{DSC}} = 1 - \frac{2\,|\hat{\mathbf{S}} \cdot \mathbf{S}|_1}{|\hat{\mathbf{S}}|_1 + |\mathbf{S}|_1}, \tag{8}$$

where $\hat{\mathbf{S}}$ and $\mathbf{S}$ denote the predicted probabilities and ground truth binary mask, respectively. Additionally, due to the sparsity of annotated regions, we employ the Weighted Cross-Entropy (WCE) loss, $\mathcal{L}_{\text{WCE}}$, to mitigate class imbalance and reduce the learning bias between foreground and background during training. The WCE loss is defined as:

$$\mathcal{L}_{\text{WCE}} = -(w_1 \cdot \mathbf{S} \cdot \log(\hat{\mathbf{S}}) + w_2 \cdot (1 - \mathbf{S}) \cdot \log(1 - \hat{\mathbf{S}})), \tag{9}$$

where $w_1 = \frac{N_{\text{total}}}{2 \times N_{\text{pos}}}$ and $w_2 = \frac{N_{\text{total}}}{2 \times N_{\text{neg}}}$ are the class weights for foreground and background voxels, respectively. These weights are computed based on the class distribution over the entire volume, where N_{total} denotes the total number of voxels, and N_{pos} and N_{neg} indicate the number of voxels labeled as foreground and background, respectively. Finally, the overall loss function is defined as:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{DSC}} + (1 - \alpha)\mathcal{L}_{\text{WCE}}, \tag{10}$$

where α is a weighting factor that balances the contributions of $\mathcal{L}_{\text{DSC}}$ and $\mathcal{L}_{\text{WCE}}$, and is determined through a grid search, with 0.6 identified as the optimal setting.

V. Experiments

A. Experimental Settings

In our proposed method, the key hyperparameters are set as follows. The initial learning rate is set to 1×10^{-4}, and the Adam optimizer was employed to minimize the loss function. A cosine annealing scheduler is used to gradually reduce the learning rate. For data augmentation, random horizontal flipping, random translations, and Gaussian smoothing were applied. The input data was cropped randomly, with sub-volumes of size $96 \times 96 \times 96$ extracted from the original volumes. To prevent redundancy, adjacent sub-volumes were ensured to have no overlap. Additionally, sub-volumes consisting entirely of background pixels were excluded from the dataset, ensuring that the network focused on the vessel structures. The batch size was set to 1, and weight decay was introduced with a coefficient of 1×10^{-5} to regularize the model and mitigate overfitting. The training process was carried out over a total of 100 epochs.

Furthermore, We compare our method with 13 state-of-the-art methods, belonging to three categories. The first category consists of 3 widely used general segmentation methods, i.e., UNet [38], UNETR [12], and nnFormer [13], as well as two CLIP driven segmentation foundation models, namely UniModel [39] and VISTA3D [40]. The second category includes 5 segmentation methods specifically designed for tubular structures in medical images, i.e., WingsNet [41], ERNet [42], Pvsvas [5], CSNet [43], and SerpMamba [31]. Moreover, we further compare 3 transfer learning methods tailored for medical images, i.e., AViT [21], UniSeg [23], and PGT [22], along with the same pre-trained backbone of our method. For fair comparison, all compared methods are implemented under the same settings as ours, including the data partitioning and pre-processing.

B. Evaluation on Pulmonary Vessel Segmentation

In this section, the pulmonary vessel segmentation experiments are conducted over our PV25 dataset and a publicly available CARVE14 dataset [10]. The PV25 dataset comprises 25 chest CT scans, while the CARVE14 dataset contains 55 scans. For the PV25 dataset, airway labels are used to compute an auxiliary loss alongside the vessel label loss. For the CARVE14 dataset, which lacks airway annotations, we fine-tuned the model using pre-trained weights from PV25, freezing the airway decoder and excluding the airway loss, and computed only the vessel label loss. It is worth noting that, to ensure the robustness and reliability of the evaluation, a five-fold cross-validation strategy was adopted for both datasets. Specifically, all data in the PV25 and CARVE14 datasets were divided into five folds, with each fold used once as the test set and the remaining folds used for training.

1) Evaluation With Full-Scale Training Data: Fig. 6 shows the qualitative experimental results of our method and the compared methods on the randomly selected test scans. Besides, the difference map between the predicted and ground-truth segmentation are also presented for better comparison. As shown, most methods are able to segment the major vessels. However, for thinner vessels, the results of the compared

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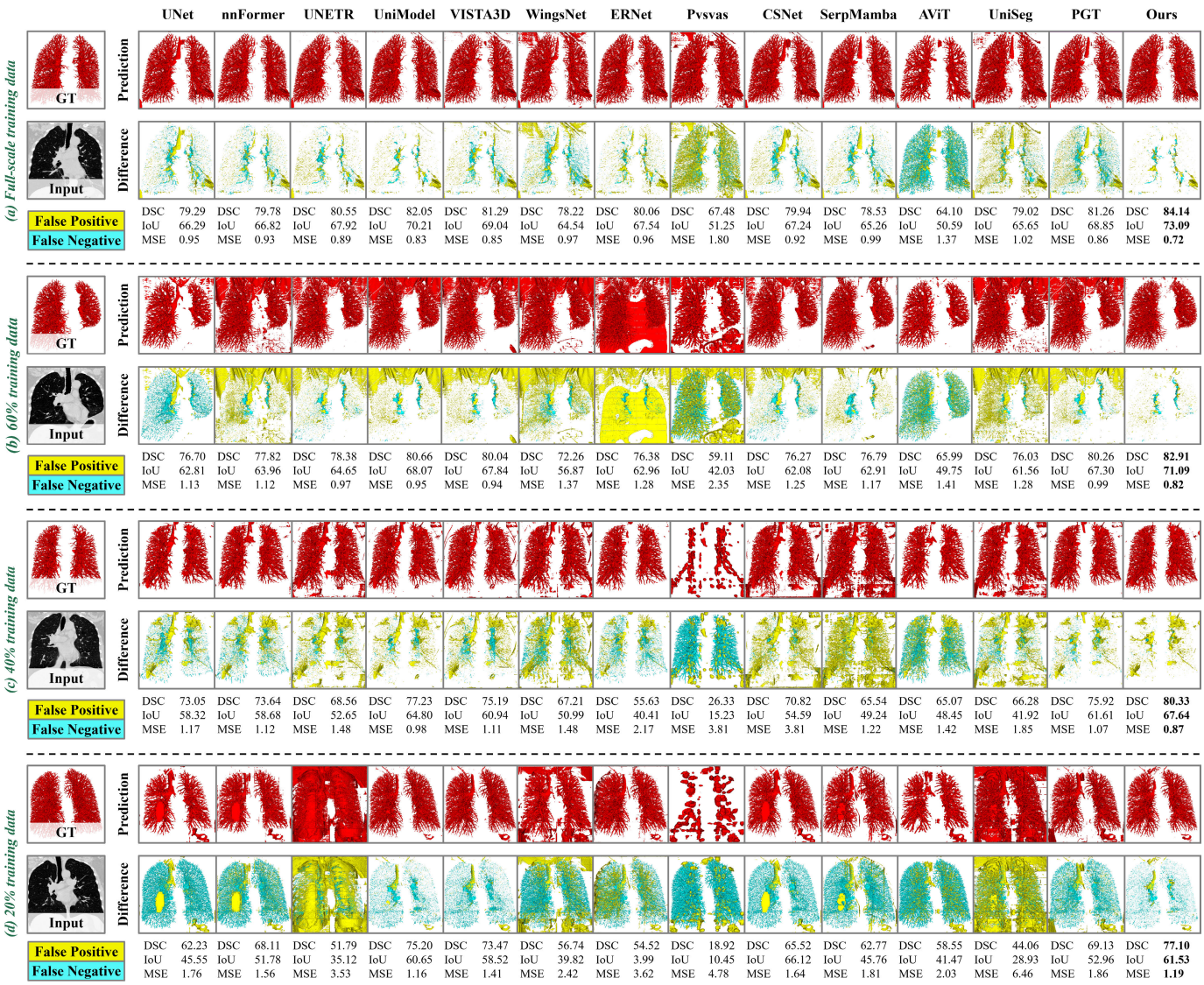


Fig. 6. The subjective results of pulmonary vessel segmentation results obtained by our method and the compared approaches under (a) full-scale, (b) 60%, (c) 40%, and (d) 20% training data settings. The difference maps between the predicted and ground-truth segmentations are also provided for better visual comparison, where yellow and turquoise regions represent false positive and false negative predictions.

methods tend to be under-segmentation or over-segmentation, as highlighted in the regions marked in yellow and turquoise in the figure. For instance, Pvsvas and AViT fail to detect tubular structures at different scales, while UniSeg tends to incorrectly detect the background as vessels. In contrast, the proposed method achieves the best segmentation results by capturing tubular topology and utilizing additional airway information, reducing both over- and under-segmentation errors. In addition to the subjective results, Table II compares the pulmonary vessel segmentation performance of our and compared methods, in terms of Dice Similarity Coefficient (DSC), centerline Dice (clDice), F1-Score, Intersection over Union (IoU), and Mean Squared Error (MSE). Note that the lower MSE indicates the better performance. Specifically, on the PV25 dataset, the proposed method achieves the best performance across all evaluation metrics, with an average DSC of 84.27%, F1-Score of 84.38%, IoU of 73.17%, and MSE of 0.76, showing notable improvements over other methods. Compared to the second best segmentation backbone, UNETR, the mean improvements in DSC, F1-Score, IoU, and MSE are approximately 2.8%, 1.91%, 1.17%, and 0.07, respectively, suggesting the effectiveness of the proposed method. Our method also demonstrates impressive performance on the CARVE14 dataset, with a DSC of 85.43%, F1-Score of 85.66%, IoU of 74.57%, and MSE of 0.46, outperforming all the compared methods. The above results verify both effectiveness and generalization capability of the proposed method. Furthermore, compared to transfer learning methods such as AViT, UniSeg and PGT, our proposed tubular-aware prompt-tuning framework demonstrates superior efficacy in transferring knowledge from the abdominal source domain to the pulmonary vessel target domain.

2) Evaluation With Limited Training Data: To evaluate the proposed method under few-shot scenarios, we conducted experiments by progressively reducing the scale of training data. Specifically, 60%, 40%, and 20% of the training sets from the PV25 and CARVE14 datasets were used for training our

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TABLE II

QUANTITATIVE COMPARISON OF PULMONARY VESSEL SEGMENTATION BETWEEN OUR METHOD AND OTHERS UNDER FULL-SCALE TRAINING SCENARIO ON THE PV25 AND CARVE14 DATASETS, IN TERMS OF DSC, CLDICE, F1-SCORE, IOU, AND MSE. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND THE SECOND-BEST RESULTS IN UNDERLINED

Category	Method	PV25									
		DSC ↑	clDice ↑	F1-Score ↑	IoU ↑	MSE ↓	DSC ↑	clDice ↑	F1-Score ↑	IoU ↑	MSE ↓
A	UNet	79.29 ± 3.27	81.43 ± 4.15	79.81 ± 4.19	66.29 ± 4.52	0.95 ± 0.23	84.08 ± 1.70	85.23 ± 2.19	84.20 ± 2.45	72.58 ± 2.53	0.52 ± 0.09
	nnFormer	79.78 ± 3.16	80.58 ± 3.54	80.49 ± 5.10	66.82 ± 4.51	0.93 ± 0.21	82.56 ± 1.58	83.63 ± 1.98	82.56 ± 2.28	70.33 ± 2.28	0.56 ± 0.09
	UNETR	80.55 ± 4.06	82.17 ± 5.32	80.95 ± 4.76	67.92 ± 5.88	0.89 ± 0.23	82.04 ± 1.94	83.55 ± 3.13	82.17 ± 2.60	69.60 ± 2.79	0.58 ± 0.10
	UniModel	82.05 ± 4.11	80.89 ± 5.24	82.54 ± 4.14	70.21 ± 6.04	0.83 ± 0.23	85.40 ± 1.62	83.93 ± 2.60	85.52 ± 2.39	74.55 ± 2.46	0.46 ± 0.06
	VISTA3D	81.29 ± 4.03	79.85 ± 3.50	81.85 ± 4.22	69.04 ± 5.81	0.85 ± 0.23	85.53 ± 1.63	83.30 ± 3.20	85.64 ± 2.39	74.75 ± 2.48	0.46 ± 0.09
B	WingsNet	78.22 ± 3.31	78.58 ± 3.54	78.86 ± 3.73	64.54 ± 4.67	0.97 ± 0.20	82.61 ± 1.94	83.15 ± 2.16	82.79 ± 2.91	70.43 ± 2.87	0.56 ± 0.10
	ERNet	80.06 ± 3.36	80.61 ± 3.41	80.89 ± 3.61	67.54 ± 4.76	0.96 ± 0.23	81.13 ± 2.14	82.24 ± 3.57	81.28 ± 2.96	68.34 ± 3.06	0.56 ± 0.08
	Pvsvas	67.48 ± 3.13	70.97 ± 4.75	68.19 ± 3.98	51.25 ± 3.46	1.80 ± 0.21	77.71 ± 2.79	81.07 ± 5.79	77.96 ± 3.53	62.98 ± 3.39	0.65 ± 0.12
	CSNet	79.94 ± 3.11	80.29 ± 3.52	80.20 ± 3.27	67.24 ± 4.32	0.92 ± 0.25	81.63 ± 1.61	81.97 ± 2.47	83.84 ± 2.02	68.92 ± 2.71	0.56 ± 0.09
	SerpMamba	78.53 ± 3.25	81.83 ± 4.11	79.20 ± 3.43	65.26 ± 4.41	0.99 ± 0.23	83.69 ± 1.74	85.65 ± 3.34	83.88 ± 1.75	72.00 ± 2.58	0.53 ± 0.09
C	UNETR + AViT	67.10 ± 1.54	63.18 ± 3.25	67.51 ± 2.68	50.59 ± 1.89	1.37 ± 0.22	71.05 ± 2.06	68.91 ± 3.71	71.10 ± 2.26	55.28 ± 2.43	0.99 ± 0.13
	UNETR + UniSeg	79.02 ± 3.57	82.04 ± 4.63	79.64 ± 3.78	65.65 ± 4.52	1.02 ± 0.19	82.56 ± 1.89	84.53 ± 3.23	82.96 ± 2.23	70.34 ± 2.47	0.58 ± 0.09
	UNETR + PGT	81.26 ± 4.14	82.67 ± 5.40	81.61 ± 3.52	68.85 ± 5.07	0.86 ± 0.22	83.38 ± 1.76	85.14 ± 2.39	83.50 ± 2.49	71.54 ± 2.57	0.53 ± 0.10
	Ours	**84.14 ± 3.38**	**85.11 ± 3.51**	**84.46 ± 4.42**	**73.09 ± 5.20**	**0.72 ± 0.20**	**86.10 ± 1.65**	**87.86 ± 2.11**	**86.21 ± 2.34**	**75.64 ± 2.53**	**0.44 ± 0.08**

A: General segmentation networks. **B**: Tubular-based segmentation models. **C**: Segmentation with Transfer Learning.

TABLE III

QUANTITATIVE COMPARISON OF OUR AND COMPARED METHODS WHEN USING 60%, 40%, AND 20% OF AVAILABLE TRAINING DATA, IN TERMS OF DSC, CLDICE, F1-SCORE, IOU, AND MSE

Training Data	Method	PV25									
		DSC ↑	clDice ↑	F1-Score ↑	IoU ↑	MSE ↓	DSC ↑	clDice ↑	F1-Score ↑	IoU ↑	MSE ↓
60%	UNETR + AViT	65.99 ± 2.41	61.40 ± 4.48	66.92 ± 3.24	49.75 ± 2.68	1.41 ± 0.24	69.76 ± 2.21	67.37 ± 2.10	69.82 ± 2.11	53.62 ± 2.60	0.90 ± 0.12
	UNETR + UniSeg	76.03 ± 1.12	77.82 ± 1.75	77.25 ± 3.10	61.56 ± 1.16	1.28 ± 0.13	78.53 ± 1.56	80.28 ± 1.45	78.71 ± 2.87	64.74 ± 2.11	0.77 ± 0.09
	UNETR + PGT	80.26 ± 2.91	77.91 ± 3.13	80.63 ± 3.40	67.30 ± 3.96	0.99 ± 0.25	81.08 ± 1.33	82.64 ± 2.76	81.24 ± 2.61	68.20 ± 1.90	0.62 ± 0.09
	Ours	**82.91 ± 2.99**	**83.61 ± 2.59**	**83.84 ± 3.24**	**71.09 ± 4.22**	**0.82 ± 0.23**	**85.77 ± 1.53**	**86.51 ± 1.87**	**85.85 ± 2.14**	**75.13 ± 2.32**	**0.44 ± 0.09**
40%	UNETR + AViT	65.07 ± 2.59	60.65 ± 4.72	66.79 ± 3.97	48.45 ± 2.73	1.42 ± 0.27	68.12 ± 2.36	63.49 ± 2.26	68.18 ± 2.21	51.71 ± 2.72	1.00 ± 0.13
	UNETR + UniSeg	66.28 ± 2.04	70.81 ± 3.96	68.38 ± 3.11	41.92 ± 2.26	1.85 ± 0.24	74.03 ± 1.51	77.67 ± 2.82	74.89 ± 2.92	60.84 ± 1.99	0.92 ± 0.06
	UNETR + PGT	75.92 ± 2.29	74.04 ± 3.51	77.68 ± 2.59	61.61 ± 3.41	1.07 ± 0.26	78.10 ± 1.11	79.06 ± 2.02	78.21 ± 2.26	65.23 ± 1.58	0.71 ± 0.10
	Ours	**80.33 ± 3.09**	**81.82 ± 3.32**	**81.45 ± 2.03**	**67.64 ± 3.79**	**0.87 ± 0.29**	**85.21 ± 0.81**	**86.63 ± 2.04**	**85.27 ± 1.61**	**74.26 ± 1.24**	**0.47 ± 0.07**
20%	UNETR + AViT	58.55 ± 3.72	51.98 ± 4.83	58.94 ± 3.81	41.47 ± 3.81	2.03 ± 0.43	62.97 ± 1.33	58.31 ± 3.41	63.01 ± 1.37	46.00 ± 1.42	1.01 ± 0.07
	UNETR + UniSeg	44.06 ± 1.28	52.04 ± 3.57	44.46 ± 1.84	28.93 ± 1.09	6.46 ± 1.67	66.20 ± 1.88	72.36 ± 2.86	66.32 ± 2.35	49.55 ± 2.07	1.33 ± 0.07
	UNETR + PGT	69.13 ± 4.21	67.78 ± 5.36	69.54 ± 4.19	52.96 ± 4.02	1.86 ± 0.42	68.92 ± 1.54	66.96 ± 4.79	69.03 ± 2.30	52.64 ± 1.82	1.10 ± 0.05
	Ours	**77.10 ± 2.19**	**78.66 ± 2.25**	**77.74 ± 2.34**	**61.53 ± 3.68**	**1.19 ± 0.20**	**84.43 ± 0.44**	**85.20 ± 1.55**	**84.66 ± 1.42**	**73.34 ± 0.67**	**0.49 ± 0.03**

method and the compared approaches. For a fair comparison, we employed five-fold cross-validation for each experimental setting to ensure the reliability of the evaluation. Table III summarizes the pulmonary vessel segmentation performance of our and the compared methods with the same segmentation backbone and different transfer learning techniques. As shown in the table, the performance of all models significantly deteriorates as the amount of training data decreases. For instance, UniSeg achieves a DSC of 80.78% on PV25 with full data, but drops to 78.89% and 61.31% with 50% and 20% of the data, respectively. Notably, in the extreme situation of using only 5% of the training data, all compared methods fail to accurately segment the pulmonary vessels. However, our method continues to perform well, achieving DSC scores of 77.36% on PV25 and 82.85% on CARVE14. Fig. 6 (d) visualizes the segmentation results when only 20% of the data is used. As shown, methods such as UNETR, WingsNet, UniSeg, PGT, and Pvsvas exhibit noticeable collapses, resulting in over-segmentation or under-segmentation artifacts. In contrast, the proposed method can still produce reasonable and accurate segmentation results when given limited training data. The above results demonstrate the robustness of our method for pulmonary vessel segmentation in few-shot scenarios, highlighting its effectiveness even when the available data is extremely limited. The tubular-aware designs can bring significantly larger improvements when the training data decreases.

C. Evaluation on External Segmentation Tasks

To further assess the effectiveness and generalization ability of the proposed method, we conduct additional experiments on two other key medical image segmentation tasks: pulmonary airway segmentation and pulmonary artery-vein segmentation. These tasks are introduced to evaluate the ability of the proposed tubular-aware designs to consistently leverage tubular features and cross-correlations, allowing anatomical priors learned from one pulmonary tubular structure to enhance the segmentation of another pulmonary tubular structure.

1) Pulmonary Airway Segmentation: Here, 300 CT scans from the ATM22 dataset are selected to evaluate the performance of different methods on the pulmonary airway segmentation task. The scans were randomly divided into training, validation, and test sets, comprising 225, 30, and

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TABLE IV
AIRWAY SEGMENTATION RESULTS OVER ATM22

Method	DSC ↑	F1-Score ↑	IoU ↑	MSE ↓
UNet	78.35 ± 1.05	78.86 ± 1.08	65.12 ± 1.76	1.72 ± 0.13
nnFormer	83.86 ± 4.35	84.38 ± 4.73	73.06 ± 6.68	1.09 ± 0.06
UNETR	87.25 ± 4.09	87.43 ± 3.98	77.62 ± 6.32	0.94 ± 0.04
UniModel	90.61 ± 4.80	90.94 ± 4.78	82.88 ± 7.38	0.71 ± 0.04
VISTA3D	90.21 ± 4.60	90.59 ± 4.13	82.51 ± 5.16	0.77 ± 0.06
WingsNet	89.76 ± 3.99	90.05 ± 4.05	81.65 ± 6.35	0.74 ± 0.04
ERNet	**91.66 ± 1.99**	91.38 ± 2.26	**84.67 ± 3.37**	0.62 ± 0.03
Pvsvas	78.21 ± 2.27	78.56 ± 2.36	64.53 ± 3.57	1.58 ± 0.06
CSNet	86.50 ± 4.51	88.05 ± 5.17	78.57 ± 6.22	0.83 ± 0.08
SerpMamba	88.65 ± 5.35	89.47 ± 4.36	79.03 ± 6.26	0.74 ± 0.06
AViT	74.18 ± 5.29	74.91 ± 5.03	59.40 ± 5.68	1.78 ± 0.08
UniSeg	88.81 ± 2.37	89.08 ± 3.18	79.98 ± 3.71	0.84 ± 0.03
PGT	82.41 ± 4.01	82.12 ± 4.16	70.25 ± 5.67	1.24 ± 0.04
Ours	91.44 ± 2.94	**91.55 ± 3.04**	84.36 ± 4.80	**0.54 ± 0.03**

TABLE V
PERFORMANCE COMPARISON OF ARTERY-VEIN SEGMENTATION ACROSS DIFFERENT METHODS

Method	ACC ↑	DSC ↑	F1-Score ↑	FPR ↓
UNet	85.58 ± 5.52	79.20 ± 6.12	79.60 ± 4.98	1.35 ± 0.53
nnFormer	84.95 ± 2.96	76.40 ± 2.33	76.40 ± 1.37	1.65 ± 0.79
UNETR	72.56 ± 4.09	65.82 ± 4.52	66.03 ± 4.06	1.87 ± 0.35
UniModel	90.65 ± 4.02	81.02 ± 4.35	82.07 ± 5.25	0.41 ± 0.07
VISTA3D	89.04 ± 3.12	80.66 ± 3.24	80.73 ± 3.47	0.40 ± 0.06
WingsNet	90.34 ± 4.66	80.27 ± 5.86	81.05 ± 6.61	0.38 ± 0.06
ERNet	90.87 ± 5.35	81.98 ± 5.86	84.50 ± 6.69	0.40 ± 0.07
Pvsvas	82.60 ± 5.33	76.59 ± 5.64	77.14 ± 5.22	0.57 ± 0.06
CSNet	88.73 ± 4.32	78.15 ± 5.26	78.36 ± 4.58	0.40 ± 0.06
SerpMamba	89.62 ± 4.11	80.02 ± 4.51	82.39 ± 5.72	0.39 ± 0.05
AViT	81.74 ± 4.73	72.41 ± 4.41	72.60 ± 4.22	1.83 ± 0.72
UniSeg	66.05 ± 3.21	59.64 ± 3.57	59.88 ± 3.05	0.64 ± 0.06
PGT	65.25 ± 3.26	61.31 ± 3.48	61.04 ± 2.43	1.98 ± 0.11
Ours	**91.22 ± 3.42**	**83.55 ± 3.57**	**84.62 ± 2.55**	**0.32 ± 0.05**

45 scans, respectively. Note that, the airway annotation of our PV25 dataset sources from ATM22 dataset. Thus, the experiments for pulmonary airway segmentation are conducted over ATM22, which has more evaluation samples. Since paired vessel labels are unavailable, only airway labels are used to compute the loss, with the vessel decoder frozen. As shown in Table IV, the proposed method obtains a DSC of 91.44%, which is on par with the state-of-the-art model, ERNet, tailored for medical image segmentation. Meanwhile, our method significantly surpasses the rest compared methods, especially general segmentation methods of UNet, nnFormer, and UNETR, by over 13.09%, 7.58%, and 4.19% in DSC, respectively. This validates that the proposed method achieves excellent cross-task generalization. The performance mainly benefits from the developed CCG module that explicitly models multi-level correlations between pulmonary vessels and airways, enabling better representation learning even across different but related anatomical structures. Compared to transfer learning-based methods, our method also demonstrates obvious advantages. For example, AViT and PGT only achieve 74.18% and 82.41% in DSC, respectively, indicating limited capability in adapting prompt-based strategies to the airway domain. UniSeg performs relatively better with a DSC of 88.81%, but still lags behind our method by 2.63%. In contrast to these methods, our approach directly leverages structural correlations, making it more robust across tasks. In addition to DSC, our method yields the highest F1-score (91.55%), competitive performance of IoU (84.36%), and the lowest MSE (0.54), demonstrating its effectiveness in capturing airways with high precision and recall.

2) Pulmonary Artery-Vein Segmentation: To further validate the effectiveness of our method, we evaluate its performance on the task of pulmonary artery-vein segmentation over CARVE14 dataset. This dataset includes 55 CT scans annotated with both arteries and veins, in which 45, 5, and 10 CT scans are allocated to the training, validation and test sets. Notably, only artery and vein labels are used to compute the loss, with the airway decoder frozen and airway loss excluded. Table V presents the performance of our and compared methods for pulmonary artery-vein segmentation. As listed, our method achieves an average accuracy (ACC) of 91.22%, a DSC of 83.55%, an F1-Score of 84.62%, and a false positive rate (FPR) of 0.32, all of which are the best results among the compared models. Specifically, compared to UNETR (the backbone model of our method), the proposed method improves 18.66%, 17.73%, 18.59%, and 1.55 in ACC, DSC, F1-Score, and FPR, respectively. The above results indicate that our method not only achieves accurate vessel segmentation but also demonstrates remarkable performance in the more challenging task of distinguishing between pulmonary arteries and veins.

TABLE VI
ABLATION STUDIES OF THE PROPOSED METHOD

Method	Components		DSC ↑	F1-Score ↑	IoU ↑	MSE ↓
	MDRG	CCG				
Baseline	×	×	80.55 ± 4.06	80.95 ± 4.76	67.92 ± 5.88	0.89 ± 0.23
Model 1	✓	×	82.97 ± 3.38	83.33 ± 4.69	72.04 ± 4.95	0.83 ± 0.24
Model 2	×	✓	83.56 ± 3.42	83.92 ± 4.45	72.85 ± 4.64	0.81 ± 0.21
Model 3	✓	✓	84.14 ± 3.38	84.46 ± 4.42	73.09 ± 5.20	0.72 ± 0.20

D. Ablation Study

1) Ablation Study of Key Components: To systematically evaluate the effectiveness of the proposed components in few-shot scenarios, we conduct an ablation study on the PV25 dataset, analyzing the contributions of the MDRG and CCG modules. The results are summarized in Table VI.

In Model 1, the integration of the MDRG module into the baseline resulted in significant improvements across all evaluation metrics. The DSC, F1-Score, IoU, and MSE increased from 81.47%, 83.47%, 72.00%, and 0.83 to 83.77%, 83.83%, 72.53%, and 0.81, respectively. These results demonstrate that the MDRG module effectively captures multi-scale tubular features while preserving vessel structure integrity, even with limited training data.

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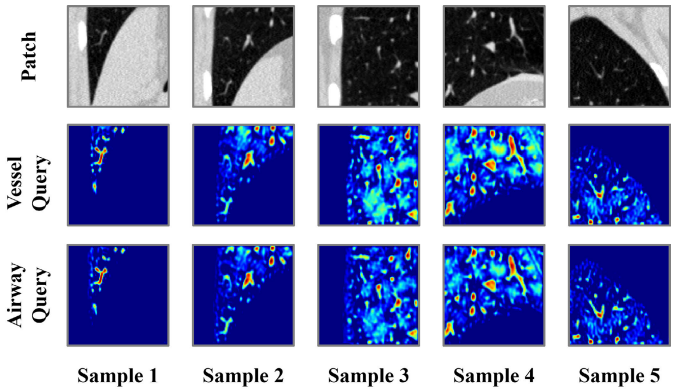


Fig. 7. Visualization of image patches and attention maps comparing the use of vessel features versus airway features as the query in the CCG module for vessel segmentation. The results demonstrate that both approaches are equally effective in focusing on the vessel regions.

TABLE VII
ABLATION STUDY COMPARING THE USE OF VESSEL AND AIRWAY FEATURES AS A QUERY IN THE CROSS-ATTENTION MECHANISM OF THE CCG MODULE FOR VESSEL SEGMENTATION. THE RESULTS INDICATE THAT BOTH CONFIGURATIONS LEAD TO COMPARABLE SEGMENTATION ACCURACY

Method	DSC ↑	F1-Score ↑	IoU ↑	MSE ↓
Vessel features as query	84.27 ± 3.41	84.42 ± 4.43	73.17 ± 5.20	0.76 ± 0.21
Airway features as query	84.31 ± 3.38	84.39 ± 4.40	73.13 ± 5.17	0.76 ± 0.19

In Model 2, we incorporated the CCG module into the baseline, utilizing multi-scale airway information to provide contextual guidance for improved pulmonary vessel segmentation. Experimental results show that the CCG module enhanced the DSC to 84.02%, improved the F1-Score to 84.38%, increased the IoU to 73.15%, and reduced the MSE to 0.79, demonstrating its effectiveness in leveraging the structural correlation between airways and vessels to boost segmentation performance.

In Model 3, we integrated both the MDRG and CCG modules into the baseline. As shown in Table VI, this combination fully leveraged the synergistic effects of both modules, achieving the best overall results. Compared to the baseline, the DSC improved by 2.8%, the F1-Score increased by 0.95%, the IoU improved by 1.17%, and the MSE decreased by 0.76.

2) Structure Comparison of CCG Module: Table VII presents the results of a comparison between using vessel and airway features as query in the cross-attention mechanism of the CCG module for vessel segmentation. The results show that both structures perform nearly similar, with only slight differences in DSC, F1-Score, IoU, and MSE, which are 84.27, 84.42, 73.17, and 0.76 for the vessel features as query and 84.31, 84.39, 73.13, and 0.76 for the airway features as query, respectively. This comparable performance can be attributed to the similarities in their branching patterns and topological structures. Consequently, both features provide valuable spatial and morphological guidance to each other. Fig. 7 displays five visual samples, showing image patches alongside the corresponding attention maps when either vessel or airway features are used as query in the CCG module. As shown, both structures effectively focus on the vessel regions (red area). This suggests that both vessel and airway features are effective as query in the CCG module.

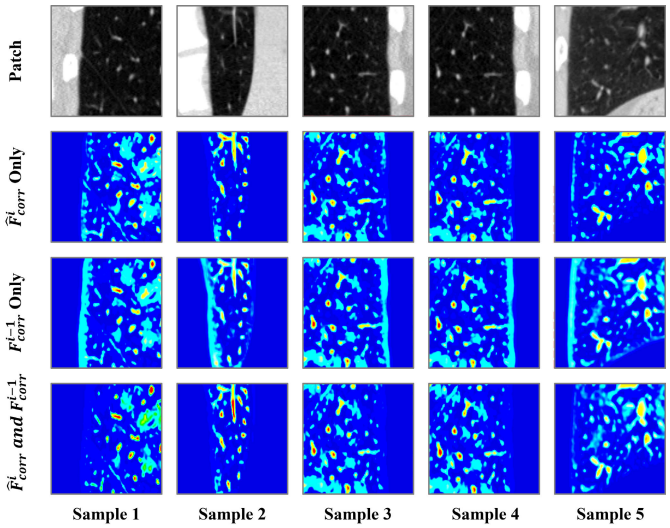


Fig. 8. Visualization of attention maps corresponding to different correlation designs in the CCG module.

TABLE VIII
AN ABLATION STUDY EVALUATING DIFFERENT CORRELATION DESIGNS IN THE CCG MODULE FOR PULMONARY VESSEL SEGMENTATION PERFORMANCE, COMPARING AIRWAY-VESSEL CORRELATIONS $\widehat{F}^{i}_{\text{CORR}}$, INTER-DECODER CORRELATIONS F^{i-1}_{CORR}, AND THEIR COMBINATION

Method	DSC ↑	F1-Score ↑	IoU ↑	MSE ↓
$\widehat{F}^{i}_{\text{corr}}$ Only	83.85 ± 3.74	84.23 ± 4.66	72.49 ± 5.30	0.78 ± 0.20
F^{i-1}_{corr} Only	83.21 ± 3.77	83.65 ± 3.55	71.97 ± 4.82	0.79 ± 0.20
$\widehat{F}^{i}_{\text{corr}}$ and F^{i-1}_{corr}	84.27 ± 3.41	84.42 ± 4.43	73.17 ± 5.20	0.76 ± 0.21

3) Effectiveness of Correlation Features in the CCG Module: We conducted ablation experiment to evaluate the effectiveness of different correlation designs in the CCG module, by only using airway-vessel correlation features ($\widehat{F}^{i}_{\text{corr}}$) or cross-layer correlation features (F^{i-1}_{corr}). Table VIII and Fig. 8 illustrate the quantitative and visualization results of this ablation study, respectively. With both airway-vessel and cross-layer correlation features, the proposed CCG module yields the best performance of 84.27 in DSC, 84.42 in F1-score, 73.17 in IoU, and 0.76 in MSE. Similarly, the visualization results in Fig. 8 show that the complete version of CCG module is able to focus more accurately on the vessels. The above results confirm that both $\widehat{F}^{i}_{\text{corr}}$ and F^{i-1}_{corr} are essential for enhancing vessel segmentation accuracy.

E. Evaluation Across Different Backbones

To further validate the transferability of our method, we replaced the UNETR [12] backbone with four representative segmentation architectures, including UNet [38], nnFormer [13], UniModel [39], and VISTA3D [40], and evaluated all models on the PV25 dataset. For a fair comparison, we kept the training protocol, optimization settings, and data splits identical to those used with the UNETR backbone. As shown in Table IX, our method consistently improves the performance

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TABLE IX
SEGMENTATION PERFORMANCE OF OUR TUBULAR-AWARE PROMPT-TUNING FRAMEWORK ACROSS DIFFERENT BACKBONE MODELS

Method	PV25			
	DSC ↑	F1-Score ↑	IoU ↑	MSE ↓
UNet	79.29 ± 3.27	79.81 ± 4.19	66.29 ± 4.52	0.95 ± 0.23
UNet + Ours	80.98 ± 3.20	81.47 ± 4.68	68.64 ± 4.57	0.88 ± 0.21
nnFormer	79.78 ± 3.16	80.49 ± 5.10	66.82 ± 4.51	0.93 ± 0.21
nnFormer + Ours	80.58 ± 3.23	81.02 ± 4.68	68.03 ± 4.60	0.89 ± 0.20
UNETR	80.55 ± 4.06	80.95 ± 4.76	67.92 ± 5.88	0.89 ± 0.23
UNETR + Ours	84.14 ± 3.38	84.46 ± 4.42	73.09 ± 5.20	0.72 ± 0.20
UniModel	82.05 ± 4.11	82.54 ± 4.14	70.21 ± 6.04	0.83 ± 0.23
UniModel + Ours	82.80 ± 4.15	83.42 ± 4.18	71.18 ± 5.70	0.79 ± 0.19
VISTA3D	81.29 ± 4.03	81.85 ± 4.22	69.04 ± 5.81	0.85 ± 0.23
VISTA3D + Ours	83.25 ± 4.82	83.89 ± 5.01	71.87 ± 5.73	0.77 ± 0.23

of all backbones. Their original DSC were 80.55, 79.29, 79.78, 82.05, and 81.29, which increased to 84.14, 80.98, 80.58, 82.80, and 83.25 after applying our method, yielding gains of 3.59, 1.69, 0.80, 0.75, and 1.96. These results demonstrate that the proposed tubular-aware prompt-tuning framework is potential to enhance various segmentation backbone models, confirming the strong transferability of our framework.

VI. DISCUSSION

Pulmonary vessel segmentation is crucial for diagnosing diseases like pulmonary embolism and COPD. While DNN-based methods have made significant strides, they rely heavily on large, high-quality annotated datasets, which are labor-intensive to create due to the complex structure of pulmonary vessels. This data scarcity limits the generalization of existing methods, particularly in few-shot learning scenarios, where model performance often deteriorates, hindering their reliability in clinical practice.

To address these challenges, this study introduces the PV25 dataset, the first dataset incorporating paired fine-grained annotations of both vessels and airways. Additionally, we propose TPNet, which leverages prior knowledge of tubular structures to enhance pulmonary vessel segmentation under few-shot conditions. TPNet comprises two novel components: the MDRG module and the Cross-Correlation Guided (CCG) module. The MDRG module enhances the model's ability to capture vessel features by exploiting tubular connectivity, while the CCG module effectively utilizes the structural correlation between airways and vessels, further improving segmentation accuracy.

We compare our method with three categories of approaches: general medical image segmentation models, tubular-specific models, and transfer learning methods. Generic models, using encoder–decoder architectures, often fail to capture fine-grained vessel details, especially in few-shot scenarios. CLIP-based methods like UniModel and VISTA3D improve global alignment but lack structure-aware priors, limiting vessel-specific feature capture in data-scarce settings. Tubular-specific models enhance vessel representation but segment vessels in isolation without anatomical relationships. Prompt-based transfer learning methods, such as AViT and UniSeg, struggle with vessel-specific adaptation, while PGT's grouped prompt strategy becomes unreliable under limited data. In contrast, our tubular-aware prompt tuning framework guides the network towards key tubular structures, preserving segmentation performance even in extreme data scarcity. As shown in Fig. 6, our method ensures structural completeness across both full-scale and few-shot scenarios, demonstrating superior robustness.

Experimental results on the PV25 and CARVE14 datasets show that TPNet excels in both full-scale and few-shot vessel segmentation, outperforming baseline methods. Under extreme conditions with only 20% of training data, TPNet maintains high accuracy, demonstrating its effectiveness in few-shot scenarios. Additionally, TPNet performs robustly in pulmonary airway and artery-vein segmentation tasks, highlighting its versatility. In addition, we assess the computational efficiency of our method compared to competing approaches, finding that it strikes a favorable balance between segmentation accuracy and computational cost, with a lightweight version offering improved efficiency. Besides, it is an interesting future work to evaluate and improve the generalization capability of our model in more scenarios such as pulmonary artery segmentation.

Despite its effectiveness in addressing few-shot pulmonary vessel segmentation, TPNet has certain limitations that warrant further investigation. One promising future direction is to incorporate more flexible and efficient learning paradigms to enhance model robustness under limited supervision. Recent studies have shown that source-free active domain adaptation can reduce dependence on source-domain data while selectively acquiring the most informative target-domain samples. This can be achieved through strategies such as uncertainty estimation, reconstruction of target-domain feature distributions, and reliable pseudo-labeling. Incorporating this paradigm into future pulmonary vessel segmentation research may allow for more efficient knowledge transfer and improved performance under extreme few-shot conditions. A complementary direction is the integration of loss functions that consider both topological and geometric properties. A topologically-informed loss preserves vessel connectivity by enforcing consistency along the centerline, while a geometrically-informed loss ensures boundary and shape fidelity, capturing subtle morphological details in small or closely spaced vessels. Incorporating a vascular-biased correlation loss can thus improve the learning of fine vessel structures and enhance robustness in few-shot scenarios.

VII. CONCLUSION

This study first introduced the PV25 dataset, the first dataset that provides paired annotations for both pulmonary vessels and airways. Additionally, we proposed TPNet, a novel method designed for few-shot pulmonary vessel segmentation. By incorporating the proposed MDRG module and CCG module, TPNet effectively leverages prior knowledge of tubular structures, enabling accurate vessel segmentation with limited

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training data. Extensive experiments demonstrate that TPNet not only achieves promising performance in pulmonary vessel segmentation but also exhibits robust performance in airway segmentation and artery-vein segmentation, highlighting its potential to serve as a competitive solution for a range of tubular structure segmentation tasks.

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